

Online Appendix: Informational Power Through Pivotality

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon + $N - 1$ weak states)	Def. 1
$\theta \in \{0, 1\}$	State of the world	Def. 1
p	Prior $\Pr(\theta = 1)$	Def. 1
μ	Posterior belief $\Pr(\theta = 1 \mid \text{history})$; equals p at entry, updated after R1 rejection	Sec. 3
$V(\theta)$	Value of cooperation: $V(0) = 1$, $V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0, 1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r - 1)$	Sec. 3
x	Shorthand $(N - 1)\alpha r$	Sec. 3
q	Majority quota $\lfloor N/2 \rfloor + 1$	Def. 1
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	R1 screening cutoff	Prop. 2
ϕ	Auxiliary for R1 cutoff: $[rN - \beta(N - 1 + r)]/[\beta(r - 1)]$	Prop. 2
$\tau(R)$	Entry threshold under rule R	Sec. 6
$v(\mu, R)$	Hegemon's net gain from institution: $E[V_H^{R1}] - \alpha V_e(\mu)$ if entry, 0 otherwise	Sec. 6
$\bar{\alpha}$	Regime threshold for α -independent R1 cutoff	Prop. 2
α^*	Threshold for conditional dominance (Thm. 1): necessary and sufficient	Eq. (4)
$\omega(\mu)$	R2 continuation shorthand: $(N - 2)\beta V_W^{R2}(\mu)$	App. A.3
κ_M	Majority weak-state coefficient: $(1 - \alpha)[N(N - 1) + \beta(q - 1)]/(N^2(N - 1))$	App. A.7
$D(\mu)$	Conditional payoff difference $V_H^{R1}(\mu, U) - V_H^{R1}(\mu, M)$	Thm. 1
Γ	Marginal value of uncertainty for unanimity	Remark 1
$\bar{\mu}$	Belief threshold above which majority dominates: $1 - D(1) /\Gamma$	Remark 1
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M	Majority payoff coefficient	App. B.1
$\mathcal{F}_R$	Formation set under rule R : $\{p : V_W^{R1}(p, R) \geq c\}$	Sec. 7

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H . The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H 's payoff is type-specific and does not depend on W 's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N-1)\alpha]}{N} \quad (1)$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \quad (2)$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W 's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad (\text{constant in } \mu), \quad (3)$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \quad (4)$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad (\text{overpaid}). \quad (5)$$

Weak states' R2 continuation values under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \quad (6)$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \quad (7)$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \quad (8)$$

$$F_1^{agg}(\mu) = (1-\mu) \left[1 - \frac{\beta(1+x)}{N} - \omega(\mu) \right] + \frac{\mu\beta r(1-\alpha)}{N} \quad (9)$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α -dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \quad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (1)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1 - \mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (3).

A.5 Case selection

The R1 screening cutoff lies on the high R2 branch ($\mu_s^{R1} > \mu_s^{R2}$, α -independent) when $\Delta_1(\mu_s^{R2}) > 0$. On the high branch, Δ_1 is a convex quadratic with $\Delta_1(0) > 0$ and $\Delta_1(1) = rN(\beta-1) < 0$, so $\Delta_1(\mu) > 0$ for all $\mu < \mu_s^{R1}$. Hence $\Delta_1(\mu_s^{R2}) > 0$ if and only if $\mu_s^{R2} < \mu_s^{R1}$. Substituting

$\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$ into $\mu_s^{R2} < \mu_s^{R1}$ and solving for α :

$$\frac{\alpha(r-1)}{r-\alpha} < \mu_s^{R1} \iff \alpha < \frac{\mu_s^{R1} \cdot r}{r-1+\mu_s^{R1}} = \bar{\alpha}(r, \beta, N),$$

which defines the threshold in equation (2).

When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch, where $V_W^{R2}(\mu) = (1-\mu)(1-\alpha)/N$ and Δ_1 depends on α through $\omega(\mu)$. The cutoff still exists and is unique: $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ hold regardless of branch, so a root exists by the intermediate value theorem; since Δ_1 is quadratic on $(0, \mu_s^{R2})$ (at most two roots) with opposite-sign endpoints, there is exactly one root. On the high branch, $\Delta_1 \leq 0$, so no additional roots exist. The jump at μ_s^{R1} remains positive. The proof of Theorem~1 (Appendix~B.5) covers both regimes explicitly: the payoff decomposition $D = D_{\text{base}} + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2} + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}$ is additive regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} , and the endpoint argument establishes $D(\mu) > 0$ on all branches in both cases.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds exactly: $V_H[1 + (N-1)\alpha]/N + (N-1)(1-\alpha)V_e(\mu)/N = V_e(\mu) \cdot [1 + (N-1)\alpha + (N-1)(1-\alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N-1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N-1)V_W \leq V_e(\mu)$.

A.7 Entry thresholds

Under majority, $V_W^{R1}(\mu, M) = \kappa_M V_e(\mu)$ where $\kappa_M = (1-\alpha)[N(N-1) + \beta(q-1)]/(N^2(N-1))$. The entry threshold is $\tau(M) = \max\{0, (c/\kappa_M - 1)/(r-1)\}$.

Under unanimity, V_W^{R1} is piecewise. On the conservative R1 branch ($\mu > \mu_s^{R1}$, which also implies $\mu > \mu_s^{R2}$ when $\alpha < \bar{\alpha}$):

$$V_W^{R1}(\mu, U) = \frac{(N+\beta)V_e(\mu) - \beta r(1+N\alpha)}{N^2}.$$

This is affine in μ . Setting $V_W^{R1} = c$ and solving:

$$\tau_U^{con} = \frac{cN^2 - [N + \beta - \beta r(1 + N\alpha)]}{(N + \beta)(r - 1)}. \quad (10)$$

On the aggressive R1 branch ($\mu < \mu_s^{R1}$), the proposer surplus F_1^{agg} is quadratic in μ (the product of $(1 - \mu)$ with the affine terms in $\omega(\mu)$ generates a second-order term). However, the full weak-state expected payoff V_W^{R1} —which averages over proposer and non-proposer roles—simplifies to a piecewise affine expression on each R2 sub-branch, because the quadratic terms in the proposer and non-proposer components cancel (see the proof of Lemma 1 in Appendix B.7). The entry threshold on each sub-branch therefore solves a linear equation.¹

The numerical results in the paper compute $\tau(U)$ by pointwise evaluation of V_W^{R1} .

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

We construct the unique PBE of the two-round Baron-Ferejohn bargaining game under majority rule, proceeding by backward induction from Round 2.

Step 1: Round 2 continuation values. Round 2 is the terminal bargaining round. If no agreement is reached, each player receives its disagreement payoff: H receives $\alpha V(\theta)$; each W_i receives 0.

Case A: H proposes in R2. The hegemon needs $q - 1$ votes from weak states. Each W_i accepts if offered $x_{W_i} \geq 0$ (its disagreement payoff). The cheapest coalition consists of $q - 1$ weak states, each offered $x_{W_i} = 0$. The hegemon keeps the entire pie:

$$V_H^{R2}(\theta, M \mid H \text{ proposes}) = V(\theta).$$

Case B: W_j proposes in R2. The proposer W_j needs $q - 1$ additional votes. Two strategies are

¹Because V_W^{R1} has a downward jump at μ_s^{R1} (opposite to V_H 's upward jump), the formation set $\{\mu : V_W^{R1} \geq c\}$ can be disconnected for intermediate values of c : entry may occur both below μ_s^{R1} (aggressive regime) and above some $\mu_2 > \mu_s^{R1}$ (conservative regime), with a gap around the cutoff. The affine formula (10) gives the entry threshold in the conservative regime and is the relevant quantity when $c > \max_{\mu \leq \mu_s^{R1}} V_W^{R1}(\mu, \text{agg})$.

available: (a) buy $q - 1$ other weak states at cost 0 each (their disagreement payoff), total cost 0; or (b) buy H at cost $\alpha V(\theta) > 0$ plus $q - 2$ other weak states at cost 0. Since $\alpha V(\theta) > 0$, strategy (a) is strictly cheaper. Hence W_j excludes H from the winning coalition.

Under strategy (a), the proposal passes with q votes. The hegemon receives its disagreement payoff $\alpha V(\theta)$; the proposer keeps $V(\theta) - \alpha V(\theta) = (1 - \alpha)V(\theta)$.

R2 continuation values (ex ante over proposer identity). Each player is recognized with probability $1/N$:

$$V_H^{R2}(\theta, M) = \frac{1}{N}V(\theta) + \frac{N-1}{N}\alpha V(\theta) = \frac{V(\theta)[1 + (N-1)\alpha]}{N}.$$

$$V_W^{R2}(\mu, M) = \frac{1}{N}(1 - \alpha)V_e(\mu) + \frac{N-2}{N} \cdot 0 + \frac{1}{N} \cdot 0 = \frac{(1 - \alpha)V_e(\mu)}{N}.$$

Step 2: Round 1 optimal offers.

Case A: H proposes in R1. The hegemon buys $q - 1$ weak states at $\beta V_W^{R2}(\mu, M)$ each and keeps the remainder:

$$\Pi_H^{\text{prop}}(\theta, \mu, M) = V(\theta) - (q - 1)\beta \cdot \frac{(1 - \alpha)V_e(\mu)}{N}.$$

Case B: W_j proposes in R1. The same exclusion logic applies. With $N - 2$ other weak states available (excluding W_j and noting $N - 2 \geq q - 1$ for $N \geq 3$), W_j can form a winning coalition without H . The cost of including H would be at least $\beta V_H^{R2}(\theta = 0, M) = \beta[1 + (N - 1)\alpha]/N > 0$, which exceeds the cost of one additional weak state $\beta V_W^{R2}(\mu, M) \geq 0$. Hence W_j excludes H , and H receives $\alpha V(\theta)$.

Step 3: Hegemon's expected R1 payoff.

$$E_\theta[V_H^{R1}(\theta, \mu, M)] = \frac{1}{N}E_\theta \left[V(\theta) - (q - 1)\beta \frac{(1 - \alpha)V_e(\mu)}{N} \right] + \frac{N-1}{N}\alpha V_e(\mu) \quad (11)$$

$$= \frac{V_e(\mu)}{N} \left[1 - \frac{(q - 1)\beta(1 - \alpha)}{N} \right] + \frac{(N - 1)\alpha V_e(\mu)}{N} \quad (12)$$

$$= \frac{V_e(\mu)}{N^2} [N - (q - 1)\beta(1 - \alpha) + N(N - 1)\alpha] \quad (13)$$

$$= \lambda_M V_e(\mu), \quad (14)$$

where

$$\lambda_M \equiv \frac{N(1 + (N - 1)\alpha) - \beta(q - 1)(1 - \alpha)}{N^2}.$$

Step 4: Affinity in μ . Since $V_e(\mu) = 1 + \mu(r - 1)$ is affine in μ and λ_M is a constant independent of μ , the product $\lambda_M V_e(\mu)$ is affine in μ .

Step 5: No screening. Coalition composition is belief-independent: in both R1 and R2, when W proposes, it forms a majority coalition from other weak states alone, regardless of μ . Because H 's vote is never needed, there is no choice between aggressive and conservative offers, no discrete switch in offer structure, and no jump in the hegemon's payoff. The screening mechanism that operates under unanimity is entirely absent.

Step 6: Structural property $\lambda_M > \alpha$.

$$\lambda_M - \alpha = \frac{N(1 + (N - 1)\alpha) - \beta(q - 1)(1 - \alpha) - N^2\alpha}{N^2} = \frac{(1 - \alpha)[N - \beta(q - 1)]}{N^2} > 0, \quad (15)$$

since $1 - \alpha > 0$ (because $\alpha < 1/r < 1$) and $N - \beta(q - 1) > 0$ (because $\beta < 1$ and $q - 1 < N$).

□

B.2 Proof of Proposition 2 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. *Existence:* $\Delta_1(0) = \beta(r - 1)/N > 0$ and $\Delta_1(1) = rN(\beta - 1) < 0$; by continuity, a root exists in $(0, 1)$.

Uniqueness and closed form under $\alpha < \bar{\alpha}$: When $\alpha < \bar{\alpha}(r, \beta, N)$, the root lies on the high R2 branch ($\mu > \mu_s^{R2}$), where Δ_1 is a convex quadratic in μ with coefficients independent of α (Appendix A.3). Since $\Delta_1(\mu_s^{R2}) > 0$ (Appendix A.5) and $\Delta_1(1) < 0$, there is exactly one root in $(\mu_s^{R2}, 1)$, given by equation (1). Below μ_s^{R2} , Δ_1 is a concave quadratic in μ : the coefficient of μ^2 is $-(N - 2)\beta(1 - \alpha)/N < 0$, arising from the product $(1 - \mu) \cdot \omega(\mu)$ in F_1^{agg} where $\omega(\mu) = (N - 2)\beta(1 - \mu)(1 - \alpha)/N$ is linear in μ . Since $\Delta_1(0) = \beta(r - 1)/N > 0$ and $\Delta_1(\mu_s^{R2}) > 0$, and a concave function that is positive at both endpoints of an interval is positive throughout, $\Delta_1(\mu) > 0$ for all $\mu \in [0, \mu_s^{R2}]$. No additional roots exist.

Uniqueness under $\alpha \geq \bar{\alpha}$: The root falls on the low R2 branch, where Δ_1 is a quadratic in μ that depends on α . On this branch, $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ (since $\mu_s^{R2} \geq \mu_s^{R1}$ on the high branch implies $\Delta_1^{\text{high}}(\mu_s^{R2}) \leq 0$, and continuity gives $\Delta_1(\mu_s^{R2}) \leq 0$). Since Δ_1 is quadratic on $(0, \mu_s^{R2})$, it has at most two roots; combined with $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$, there is exactly one root in $(0, \mu_s^{R2}]$. On the high branch, $\Delta_1 \leq 0$ for $\mu \in [\mu_s^{R2}, 1]$, so no additional roots exist there. $\square$

B.3 Proof of Proposition 3 (Jump)

Direct computation from the difference in H's R1 payoff between the conservative and aggressive branches. See Appendix A.4. $\square$

B.5a Derivation of the payoff decomposition

This appendix derives the decomposition $D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu)$ from the primitive payoff formulas in Appendices A.1–A.3.

Step 0: Primitive payoff components. The first task is to write down the hegemon's R1 payoff from its primitive components. Because proposals are random, the payoff splits naturally by proposer identity, and this separation will prove essential: each component turns out to depend on a different screening regime, which is ultimately why the two corrections are additive.

The hegemon's expected R1 payoff $E[V_H^{R1}(\mu, U)]$ has two components, corresponding to the identity of the R1 proposer.

H proposes (probability $1/N$). The hegemon offers each weak state $\beta V_W^{R2}(\mu)$ (its discounted R2 continuation) and keeps the remainder. The expected payoff across types is:

$$\Pi_H^{\text{prop}}(\mu) = \frac{V_e(\mu) - (N-1)\beta V_W^{R2}(\mu)}{N}. \quad (16)$$

W proposes (probability $(N-1)/N$). The proposing weak state makes an offer to H that depends on the R1 screening regime. Under an aggressive R1 offer ($\mu < \mu_s^{R1}$), only $\theta = 0$ accepts; $\theta = 1$ rejects and the game proceeds to R2 with posterior $\mu' = 1$. The R1 offer to H is $\beta V_H^{R2}(\theta=0, \mu'=1)$,

and $\theta = 1$'s payoff from rejection is $\beta V_H^{R2}(\theta=1, \mu'=1)$. Since $\mu' = 1 > \mu_s^{R2}$, both R2 values are evaluated in the conservative R2 regime (Appendix A.2, equations (3)–(5)):

$$\beta V_H^{R2}(\theta=0, \mu'=1) = \frac{\beta(1+x)}{N}, \quad (17)$$

$$\beta V_H^{R2}(\theta=1, \mu'=1) = \frac{\beta(r+x)}{N}. \quad (18)$$

Under a conservative R1 offer ($\mu > \mu_s^{R1}$), both types accept. The offer is set at $\beta V_H^{R2}(\theta=1, \mu'=1) = \beta(r+x)/N$ to satisfy the high type's participation constraint. Hence the expected payoff from W 's proposal is:

$$\Pi_H^W(\mu) = \frac{N-1}{N} \begin{cases} \mu \frac{\beta(r+x)}{N} + (1-\mu) \frac{\beta(1+x)}{N} & \text{if } \mu < \mu_s^{R1} \text{ (aggressive R1),} \\ \frac{\beta(r+x)}{N} & \text{if } \mu > \mu_s^{R1} \text{ (conservative R1).} \end{cases} \quad (19)$$

The total hegemon payoff is $E[V_H^{R1}(\mu, U)] = \Pi_H^{\text{prop}}(\mu) + \Pi_H^W(\mu)$.

Two features are important. First, the R1 offers to H in (17)–(18) are evaluated at the off-path posterior $\mu' = 1$ and therefore do not depend on $V_W^{R2}(\mu)$. Second, H 's proposal payoff (16) depends on $V_W^{R2}(\mu)$ but is the same regardless of the R1 regime. These two observations establish that the R1 and R2 corrections are independent.

Step 1: Benchmark payoff (aggressive R1, conservative R2). I now compute the hegemon's total payoff in a benchmark regime—aggressive R1 screening and conservative R2 screening—which serves as the reference point against which the two corrections will be measured. The choice of benchmark is not arbitrary: under this regime, both components of the payoff take their simplest form, and the corrections for each alternative regime can then be isolated one at a time.

Under aggressive R1 and conservative R2 ($V_W^{R2} = [V_e(\mu) - \alpha r]/N$), the two components are:

$$\Pi_H^{\text{prop}}(\mu)|_{\text{con-R2}} = \frac{V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]/N}{N} = \frac{N V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]}{N^2}, \quad (20)$$

$$\Pi_H^W(\mu)|_{\text{agg-R1}} = \frac{(N-1)\beta}{N} \cdot \frac{V_e(\mu) + x}{N} = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}. \quad (21)$$

Summing:

$$\begin{aligned} E_{\text{bench}}(\mu) &= \frac{N V_e(\mu) - (N-1)\beta V_e(\mu) + (N-1)\beta \alpha r + (N-1)\beta V_e(\mu) + (N-1)\beta x}{N^2} \\ &= \frac{N V_e(\mu) + (N-1)\beta(\alpha r + x)}{N^2} = \frac{N V_e(\mu) + (N-1)\beta N \alpha r}{N^2}, \end{aligned} \quad (22)$$

where the last equality uses $\alpha r + x = \alpha r + (N-1)\alpha r = N\alpha r$.

Step 2: R1 correction δ_{R1} . The next step is to compute the payoff change when the R1 regime switches from aggressive to conservative. Because this switch affects only the R1 offers that W makes to H —and not H 's own proposal, which depends on V_W^{R2} —the correction will appear exclusively in Π_H^W , confirming that it can be treated independently of the R2 regime.

When the R1 regime switches from aggressive to conservative ($\mu > \mu_s^{R1}$), Π_H^{prop} is unchanged (it depends only on V_W^{R2} , not on the R1 regime). The change is entirely in Π_H^W . Under aggressive R1:

$$\Pi_H^W|_{\text{agg}} = \frac{(N-1)\beta}{N^2} [\mu(r+x) + (1-\mu)(1+x)] = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}.$$

Under conservative R1:

$$\Pi_H^W|_{\text{con}} = \frac{(N-1)\beta(r+x)}{N^2}.$$

The correction is:

$$\delta_{R1}(\mu) \equiv \Pi_H^W|_{\text{con}} - \Pi_H^W|_{\text{agg}} = \frac{(N-1)\beta}{N^2} [(r+x) - V_e(\mu) - x] = \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \quad (23)$$

This is non-negative for all $\mu \leq 1$, with equality only at $\mu = 1$. The correction reflects the overpayment that the low type receives under conservative R1 offers.

Step 3: R2 correction δ_{R2} . I now derive the payoff change when the R2 regime switches from conservative to aggressive. By an argument symmetric to Step 2, this switch affects only H 's proposal payoff Π_H^{prop} —the R1 offers to H are pinned at the off-path posterior $\mu' = 1$, which always lies in the conservative R2 regime regardless of the on-path belief μ .

When the R2 regime switches from conservative to aggressive ($\mu < \mu_s^{R2}$), Π_H^W is unchanged (the R1 offers to H are evaluated at $\mu' = 1$, which is always in the conservative R2 regime). The

change is entirely in Π_H^{prop} . Substituting $V_W^{R2, \text{agg}}(\mu) = (1 - \mu)(1 - \alpha)/N$ (Appendix A.2) in place of $V_W^{R2, \text{con}}(\mu) = [V_e(\mu) - \alpha r]/N$:

$$\begin{aligned}\delta_{R2}(\mu) &\equiv \Pi_H^{\text{prop}}|_{\text{agg-R2}} - \Pi_H^{\text{prop}}|_{\text{con-R2}} \\ &= \frac{-(N-1)\beta}{N^2} \left[(1 - \mu)(1 - \alpha) - V_e(\mu) + \alpha r \right] \\ &= \frac{(N-1)\beta}{N^2} \left[V_e(\mu) - \alpha r - (1 - \mu)(1 - \alpha) \right].\end{aligned}\tag{24}$$

Expanding the bracket:

$$V_e(\mu) - \alpha r - (1 - \mu)(1 - \alpha) = \mu(r - \alpha) - \alpha(r - 1),$$

so $\delta_{R2}(\mu) = (N-1)\beta[\mu(r - \alpha) - \alpha(r - 1)]/N^2$, which matches the formula in Appendix B.5. At $\mu = \mu_s^{R2} = \alpha(r - 1)/(r - \alpha)$, direct substitution gives $\delta_{R2}(\mu_s^{R2}) = 0$, confirming continuity at the R2 cutoff.

Step 4: Additivity. With both corrections in hand, I can now establish that they combine additively. This is the key structural result: it means the full payoff on any branch of the belief space can be written as the benchmark plus whichever corrections are active, without interaction terms.

The two corrections are independent because they affect disjoint components of H 's payoff:

- δ_{R2} operates through Π_H^{prop} (equation (16)), which depends on $V_W^{R2}(\mu)$ but is invariant to the R1 regime.
- δ_{R1} operates through Π_H^W (equation (19)), which depends on the R1 offer type but is invariant to $V_W^{R2}(\mu)$ —because the R1 offers to H are pinned at $\beta V_H^{R2}(\theta, \mu' = 1)$, which is always evaluated at $\mu' = 1 > \mu_s^{R2}$.

Hence the full unanimity payoff is:

$$E[V_H^{R1}(\mu, U)] = E_{\text{bench}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

Step 5: Subtracting majority and collecting terms. The final step subtracts the majority payoff from the unanimity benchmark to obtain D_{base} . This completes the decomposition: the payoff

difference $D(\mu)$ on any branch equals D_{base} plus the active corrections, and all three components are affine in μ .

Under majority (Appendix A.1), $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1 + (N - 1)\alpha) - C_{\text{buy}}]/N^2$ with $C_{\text{buy}} = \beta(q - 1)(1 - \alpha)$ and $q = \lfloor N/2 \rfloor + 1$. Define $C_{\text{out}} = N(N - 1)\alpha$. Then:

$$\begin{aligned}
D_{\text{base}}(\mu) &\equiv E_{\text{bench}}(\mu) - \lambda_M V_e(\mu) \\
&= \frac{N V_e(\mu) + (N - 1)\beta N \alpha r}{N^2} - \frac{N(1 + (N - 1)\alpha) - C_{\text{buy}}}{N^2} V_e(\mu) \\
&= \frac{V_e(\mu)[N - N - C_{\text{out}} + C_{\text{buy}}] + C_{\text{out}}\beta r}{N^2} \\
&= \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \tag{25}
\end{aligned}$$

where the third line uses $N(1 + (N - 1)\alpha) = N + C_{\text{out}}$ and $(N - 1)\beta N \alpha r = C_{\text{out}}\beta r$.

The payoff difference is therefore:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu),$$

with D_{base} , δ_{R2} , δ_{R1} as in equations (25), (24), (23). $\square$

B.5 Proof of Theorem 1 (Conditional payoff dominance)

Proof roadmap. The proof compares the conditional payoff difference $D(\mu) \equiv E[V_H(\mu, U)] - E[V_H(\mu, M)]$. Under majority, the hegemon's payoff is affine in beliefs: weak proposers can exclude the hegemon, so its private information creates no screening term. Under unanimity, the payoff difference decomposes additively into three components—a baseline that captures the cost of buying agreement under each rule, and two corrections that capture the screening rent when weak proposers switch to conservative offers and the continuation-value effect of screening in the second round. Each component is affine in μ on its domain, so positivity on any branch reduces to checking endpoints. The tightest case is $\mu = 1$, where both screening corrections vanish and the comparison reduces to vote-buying and outside-option terms alone. Evaluating $D(1) > 0$ yields exactly the condition $\alpha < \alpha^*(N, \beta)$. At all other branch endpoints, the baseline term is large enough to absorb

any negative correction, so the affine structure implies $D(\mu) > 0$ for every $\mu \in (0, 1]$.

The condition is therefore pinned down by the no-uncertainty endpoint. At $\mu = 1$, the hegemon is known to be the high type, so unanimity can dominate majority only if its bargaining advantage exceeds the cost of buying all votes. The intuition for why the advantage holds throughout the belief space has two parts. When the hegemon proposes, unanimity depresses weak states' continuation values (because rejection leads to a game where the hegemon is pivotal), allowing the hegemon to buy agreement more cheaply. When a weak state proposes, unanimity forces the proposer to secure the hegemon's vote, generating the screening rent identified in Proposition 3. Both forces favor the hegemon.

The payoff difference $D(\mu)$ decomposes into a baseline affine term plus two correction terms, each itself affine. Positivity therefore reduces to checking endpoints.

Let $C_{\text{buy}} \equiv \beta(q - 1)(1 - \alpha)$ (the vote-buying cost under majority) and $C_{\text{out}} \equiv N(N - 1)\alpha$ (the aggregate outside-option cost). Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1 + (N - 1)\alpha) - C_{\text{buy}}]/N^2$.

Step 1: Decomposition. The decomposition has three components, each affine in μ . Because an affine function that is positive at both endpoints of an interval is positive throughout, it suffices to check a finite number of boundary values.

Define:

$$D_{\text{base}}(\mu) \equiv \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \quad (26)$$

$$\delta_{R2}(\mu) \equiv \frac{(N - 1)\beta[\mu(r - \alpha) - \alpha(r - 1)]}{N^2}, \quad (27)$$

$$\delta_{R1}(\mu) \equiv \frac{(N - 1)\beta(r - 1)(1 - \mu)}{N^2}. \quad (28)$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

This decomposition holds regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} . The two corrections

are additive because they affect independent components of H 's payoff: δ_{R2} captures the change in $V_W^{R2}(\mu)$ (which enters through H 's proposal offer), while δ_{R1} captures the change in the R1 offer H receives when W proposes (which depends on the R1 regime but not on V_W^{R2}).

D_{base} is the difference under aggressive R1 and conservative R2. The R2 correction vanishes at the screening cutoff: $\delta_{R2}(\mu_s^{R2}) = 0$, ensuring continuity at μ_s^{R2} . The R1 correction satisfies $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

The proof strategy is as follows. D_{base} is positive everywhere on $[0, 1]$ (Step 2). The R2 correction δ_{R2} is active only below μ_s^{R2} ; it is negative at $\mu = 0$ but vanishes at μ_s^{R2} , so the tighter endpoint condition is $D_I(0) = D_{\text{base}}(0) + \delta_{R2}(0) > 0$. The R1 correction δ_{R1} is active only above μ_s^{R1} and is non-negative throughout, so it can only help. Step 3 assembles these facts branch by branch.

Step 2: $D_{\text{base}} > 0$ on $[0, 1]$. Since D_{base} is affine in μ —it depends on μ only through $V_e(\mu) = 1 + \mu(r - 1)$, which is itself affine—checking positivity at the two endpoints $\mu = 0$ and $\mu = 1$ is both necessary and sufficient.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]}{N^2}.$$

Setting $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$: Define $D_I(\mu) \equiv D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, the payoff difference on the low-belief branch. We check $D_I(0)$ rather than $D_{\text{base}}(0)$ directly, because $\delta_{R2}(0) < 0$ —so $D_I(0)$ is the tighter endpoint condition. Write $D_I(0) = [\beta(q - 1) - \alpha d_0]/N^2$ where $d_0 = N(N - 1) - \beta((N - 1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q - 1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q - 1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N - 1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N - 1)^2(r - 1) > 0$). Hence $\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > D_I(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly. The strategy is to partition the belief space $(0, 1]$ by the two screening cutoffs and verify $D > 0$ on each branch, using the endpoint results from Steps 1–2. On every branch, D is a sum of affine functions and is therefore itself affine, so positivity at branch endpoints implies

positivity throughout.

The standard case is $\mu_s^{R2} \leq \mu_s^{R1}$, which partitions $(0, 1]$ into three branches.

Low-belief branch ($\mu < \mu_s^{R2}$): $D = D_I(\mu)$, which is affine. At the left endpoint, $D_I(0) > 0$ (Step 2). At the right endpoint, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2). Affine with positive endpoints implies $D > 0$ throughout.

Middle branch ($\mu_s^{R2} \leq \mu \leq \mu_s^{R1}$): $D = D_{\text{base}} > 0$ directly from Step 2.

High-belief branch ($\mu > \mu_s^{R1}$): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Alternative case ($\mu_s^{R1} < \mu_s^{R2}$). The additive decomposition remains valid because the two corrections affect independent components of H 's payoff (Step 1). The partition of $(0, 1]$ now has three branches:

Low-belief branch ($\mu < \mu_s^{R1}$): aggressive R1, aggressive R2. $D = D_I(\mu) = D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, which is affine. At $\mu = 0$, $D_I(0) > 0$ (Step 2). At $\mu = \mu_s^{R1}$, D_I is continuous from the left and $D_I(\mu_s^{R1}) > 0$ because D_I is affine with $D_I(0) > 0$ and $D_I(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2), and $\mu_s^{R1} < \mu_s^{R2}$.

Middle branch ($\mu_s^{R1} \leq \mu < \mu_s^{R2}$): conservative R1, aggressive R2. Both corrections are active: $D = D_{\text{base}} + \delta_{R2} + \delta_{R1}$. This is affine. At μ_s^{R1} , D jumps up by $\delta_{R1}(\mu_s^{R1}) > 0$ relative to the low branch, so $D(\mu_s^{R1}) = D_I(\mu_s^{R1}) + \delta_{R1}(\mu_s^{R1}) > D_I(\mu_s^{R1}) > 0$. At μ_s^{R2} , $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) + \delta_{R1}(\mu_s^{R2}) \geq D_{\text{base}}(\mu_s^{R2}) > 0$. Affine with positive values at both endpoints implies $D > 0$ throughout.

High-belief branch ($\mu \geq \mu_s^{R2}$): conservative R1, conservative R2. $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Combining all branches in either case: $D(\mu) > 0$ for every $\mu \in (0, 1]$. This establishes sufficiency.

Step 4: Necessity. To establish that $\alpha < \alpha^*$ is not merely sufficient but also necessary, I show that the inequality $D(\mu) > 0$ fails at $\mu = 1$ whenever $\alpha \geq \alpha^*$.

At $\mu = 1$, both corrections vanish ($\delta_{R1}(1) = 0$ and $\mu = 1 > \mu_s^{R2}$), so $D(1) = D_{\text{base}}(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2$. Since α^* is the root of $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$, $\alpha \geq \alpha^*$ implies $D(1) \leq 0$, and the strict inequality fails. $\square$

B.6 Proof of Corollary (Unanimity dominance on the formation set)

Fix $p \in \mathcal{F}_U$, so $V_W^{R1}(p, U) \geq c$. We first show that $p \in \mathcal{F}_M$. Under majority, the budget identity holds exactly: $E[V_H(\mu, M)] + (N-1)V_W(\mu, M) = V_e(\mu)$. Under unanimity, surplus destruction from discounting on the aggressive branch yields $E[V_H(\mu, U)] + (N-1)V_W(\mu, U) \leq V_e(\mu)$. Since Theorem 1 gives $E[V_H(\mu, U)] > E[V_H(\mu, M)]$, it follows that $V_W(\mu, U) < V_W(\mu, M)$ for every $\mu \in (0, 1]$. Hence $V_W^{R1}(p, M) > V_W^{R1}(p, U) \geq c$, so entry occurs under majority at p . In particular, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Since the institution forms under both rules at p , the hegemon's payoff under each rule is the expected bargaining payoff: $V_H(R, p) = E_\theta[V_H^{R1}(\theta, p, R)]$. By Theorem 1, $V_H(U, p) > V_H(M, p)$. $\square$

B.7 Lemma: Global maximum of V_W^{R1} under unanimity

Lemma 1 (Global maximum of weak-state payoff). $V_W^{R1}(\mu, U) \leq V_W^{R1}(1, U) = r(1 - \beta\alpha)/N$ for every $\mu \in (0, 1]$, with equality only at $\mu = 1$. Hence if $\mathcal{F}_U \neq \emptyset$, then $1 \in \mathcal{F}_U$.

Proof. At $\mu = 1$, the equilibrium R1 offer is conservative: $F_1^{con}(1) = r - \beta(r+x)/N - \omega(1) > 0$ while $F_1^{agg}(1) = \beta r(1 - \alpha)/N > 0$, and the conservative surplus exceeds the aggressive surplus because $\beta < 1$ implies the full pie exceeds the discounted R2 value. Hence $\bar{V}_W \equiv V_W^{R1}(1, U) = r(1 - \beta\alpha)/N$.

We bound every feasible R1 payoff candidate by $\bar{V}_W$, using the full ex ante weak-state payoff (not only the proposer surplus). Let $W_2^L(\mu) = (1 - \mu)(1 - \alpha)/N$ and $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ denote the R2 continuation values on the low and high branches, $W_2^1 = r(1 - \alpha)/N$ the R2 value at $\mu = 1$, and $x = (N-1)\alpha r$. Under a conservative R1 offer (all types accept), a non-proposing weak state receives $\beta W_2^j(\mu)$ whenever another player proposes, giving $V_W^{Cj}(\mu) = F_1^{con,j}(\mu)/N + (N-1)\beta W_2^j(\mu)/N$. Under an aggressive R1 offer ($\theta = 1$ rejects, game proceeds to R2 with posterior one), $V_W^{Aj}(\mu) = F_1^{agg,j}(\mu)/N + \beta W_2^j(\mu)/N + (N-2)[(1 - \mu)\beta W_2^j(\mu) + \mu\beta W_2^1]/N$. The equilibrium payoff $V_W^{R1}(\mu, U)$ is one of the four candidates V_W^{CH} , V_W^{AH} , V_W^{CL} , V_W^{AL} , depending on the R1 offer type and the R2 branch.

Conservative R1, high R2 ($\mu \geq \mu_s^{R2}$): $V_W^{CH}(\mu) = [(N + \beta)V_e(\mu) - \beta r(1 + N\alpha)]/N^2$, so

$\bar{V}_W - V_W^{CH}(\mu) = (N + \beta)(r - 1)(1 - \mu)/N^2 \geq 0$, with equality only at $\mu = 1$.

Aggressive R1, high R2 ($\mu \geq \mu_s^{R2}$): The proposer surplus $F_1^{agg,H}$ contains a term $(1 - \mu)\beta W_2^H(\mu)$ that is quadratic in μ , but this term cancels exactly with the corresponding non-proposer term $(1 - \mu)\beta W_2^H(\mu)$ in V_W^{AH} , leaving a result that is affine. After substituting $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ and simplifying: $\bar{V}_W - V_W^{AH}(\mu) = [(1 - \mu)(r - 1) + \mu r(1 - \beta)]/N > 0$ for all $\mu \in [0, 1]$.

Conservative R1, low R2 ($\mu < \mu_s^{R2}$): Using $W_2^L(\mu) = (1 - \mu)(1 - \alpha)/N$, $\bar{V}_W - V_W^{CL}(\mu)$ is affine in μ . At $\mu = 0$: $(r - 1)(N - \alpha\beta + \beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r - 1)(1 - \alpha)(N + \beta)/[N^2(r - \alpha)] > 0$. Affine with positive endpoints implies $\bar{V}_W - V_W^{CL} > 0$ on $[0, \mu_s^{R2}]$.

Aggressive R1, low R2 ($\mu < \mu_s^{R2}$): The same quadratic cancellation as in the high-R2 case applies: the $(1 - \mu)\beta W_2^L(\mu)$ terms in the proposer and non-proposer components cancel, and $\bar{V}_W - V_W^{AL}(\mu)$ is affine. At $\mu = 0$: $(r - 1)(N - \alpha\beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r - 1)(1 - \alpha\beta)/[N(r - \alpha)] > 0$. Hence $\bar{V}_W - V_W^{AL} > 0$ on $[0, \mu_s^{R2}]$.

Since every payoff candidate is strictly below $\bar{V}_W$ for $\mu < 1$, and the equilibrium payoff selects one of these candidates, $V_W^{R1}(\mu, U) < \bar{V}_W$ for all $\mu < 1$. The corollary follows: if $V_W^{R1}(\mu', U) \geq c$ for some μ' , then $V_W^{R1}(1, U) \geq V_W^{R1}(\mu', U) \geq c$, so $1 \in \mathcal{F}_U$. $\square$

B.8 Proof of Proposition 4 (Institutional classification)

We verify each case exhaustively. Since $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Corollary, Appendix B.6), the three cases $p \in \mathcal{F}_U$, $p \in \mathcal{F}_M \setminus \mathcal{F}_U$, and $p \notin \mathcal{F}_M$ partition $(0, 1]$.

Case (i): $p \in \mathcal{F}_U$. The institution forms under both rules. By Corollary~1, $V_H(U, p) > V_H(M, p)$, so $U \succ M$.

Case (ii): $p \in \mathcal{F}_M \setminus \mathcal{F}_U$. The institution forms under majority but not under unanimity. Under unanimity, since $p \notin \mathcal{F}_U$, the institution does not form and the hegemon receives its bilateral outside option: $V_H(U, p) = \alpha V_e(p)$. Under majority, since $p \in \mathcal{F}_M$, the institution forms and, by Proposition 1, $V_H(M, p) = \lambda_M V_e(p)$. Since $\lambda_M > \alpha$ (Appendix B.1, Step 6) and $V_e(p) > 0$, we have $V_H(M, p) = \lambda_M V_e(p) > \alpha V_e(p) = V_H(U, p)$. Hence $M \succ U$.

Case (iii): $p \notin \mathcal{F}_M$. The institution does not form under either rule (since $\mathcal{F}_U \subseteq \mathcal{F}_M$, $p \notin \mathcal{F}_M$

implies $p \notin \mathcal{F}_U$). Both rules yield the outside-option payoff $\alpha V_e(p)$, so $U \sim M$. $\square$

Appendix C: Continuous Types

This appendix extends the model to continuous types. Section C.1 establishes that the screening rent is strictly positive for any continuous distribution with full support—a distribution-free result. Sections C.2–C.4 derive closed-form payoff expressions for the uniform distribution and show that the dominance threshold α_{cont}^* weakly exceeds the binary-type threshold α^* .

C.1 Screening rent for general distributions

The model is identical to the main model except that θ is drawn from a continuous distribution F on $[1, r]$, $r > 1$, with full support and positive density f on $(1, r)$. The cooperation value is $V(\theta) = \theta$, and the outside-option constraint becomes $\alpha \in (0, 1/r)$.

Proposition 1 (Screening rent under continuous types). *(a) In R2, W 's optimal screening cutoff satisfies $\theta_2^* > 1$, and H 's screening rent is*

$$\text{Rent}_{R2} = \int_1^{\theta_2^*} \alpha(\theta_2^* - \theta) dF(\theta) > 0.$$

(b) In R1, H 's screening rent from W 's proposal is strictly positive. (c) Under majority, W 's proposals generate zero screening rent in both rounds.

Proof. R2 under unanimity. When W proposes in R2, W sets $y_H = \alpha\theta^*$ for some cutoff $\theta^* \in [1, r]$; types $\theta \leq \theta^*$ accept and types $\theta > \theta^*$ reject. W 's expected payoff is $\pi_W(\theta^*) = \int_1^{\theta^*} (\theta - \alpha\theta^*) dF(\theta)$.

Optimal cutoff $\theta_2^ > 1$.* Since $\pi_W(1) = 0$ and, for small $\epsilon > 0$, $\pi_W(1 + \epsilon) \geq (1 - \alpha(1 + \epsilon))F(1 + \epsilon) > 0$ (using $\alpha < 1/r < 1$ and full support), the global maximizer cannot be at $\theta^* = 1$.

Screening rent. H 's expected payoff from W 's R2 proposal is $\int_1^{\theta_2^*} \alpha\theta_2^* dF(\theta) + \int_{\theta_2^*}^r \alpha\theta dF(\theta)$. The

rent relative to the perfect-discrimination benchmark $\int_1^r \alpha \theta dF(\theta)$ is

$$\text{Rent}_{R2} = \int_1^{\theta_2^*} \alpha(\theta_2^* - \theta) dF(\theta) > 0.$$

Positivity: the integrand $\alpha(\theta_2^* - \theta) > 0$ for $\theta \in [1, \theta_2^*)$, and F assigns positive mass to $[1, \theta_2^*)$ since $\theta_2^* > 1$ and F has full support.

R1 under unanimity. $V_H^{R2}(\theta, U)$ is strictly increasing in θ : the H -proposes component (θ/N) is strictly increasing, and the W -proposes component is non-decreasing. In R1, W offers y_H and type θ accepts iff $y_H \geq \beta V_H^{R2}(\theta, U)$. Strict monotonicity yields a cutoff θ_1^* , and the same argument as R2 gives $\theta_1^* > 1$ and

$$\text{Rent}_{R1} = \int_1^{\theta_1^*} [\beta V_H^{R2}(\theta_1^*, U) - \beta V_H^{R2}(\theta, U)] dF(\theta) > 0.$$

Majority. Under majority, W excludes H from the winning coalition. H receives $\alpha\theta$ type by type—the perfect-discrimination benchmark—so the screening rent is zero in both rounds. $\square$

C.2 Uniform distribution: R2 equilibrium

For the remainder of this appendix, let $F = \text{Uniform}[1, r]$, so $f(\theta) = 1/(r - 1)$, $F(\theta) = (\theta - 1)/(r - 1)$, and $E[\theta] = (1 + r)/2$. Define $x \equiv (N - 1)\alpha r$.

R2 under unanimity: $\theta_2^* = r$. W 's profit derivative is $\pi'_W(\theta^*) = [(1 - 2\alpha)\theta^* + \alpha]/(r - 1)$. The root $\theta^* = \alpha/(2\alpha - 1)$ satisfies $\theta^* \leq r$ only if $(r - 1)^2 \leq 0$, which is impossible for $r > 1$. Hence $\pi'_W > 0$ on $[1, r]$ and $\theta_2^* = r$: W pools all types with the offer $y_H = \alpha r$. The same argument applies to any truncation $\text{Uniform}[a, r]$ with $a \geq 1$, so $\theta_2^* = r$ holds regardless of the R1 screening cutoff.

R2 continuation values:

$$V_H^{R2}(\theta, U) = \frac{\theta + x}{N}, \quad V_W^{R2}(\theta, U) = \frac{\theta - \alpha r}{N}, \quad (29)$$

$$V_H^{R2}(\theta, M) = \frac{\theta(1 + (N - 1)\alpha)}{N}, \quad V_W^{R2}(\theta, M) = \frac{(1 - \alpha)\theta}{N}. \quad (30)$$

C.3 R1 equilibrium

H proposes under unanimity (probability $1/N$). H offers each W its discounted R2 continuation value $\beta E_\theta[V_W^{R2}(\theta, U)] = \beta(1 + r - 2\alpha r)/(2N)$ and keeps

$$V_H^{R1}(\theta, H\text{-prop}, U) = \theta - \frac{(N-1)\beta(1+r-2\alpha r)}{2N}. \quad (31)$$

W proposes under unanimity (probability $(N-1)/N$). The proposer W_k chooses a screening cutoff θ_1^* : types $\theta \leq \theta_1^*$ accept the offer $y_H = \beta(\theta_1^* + x)/N$; types $\theta > \theta_1^*$ reject, and R2 begins with posterior $\text{Uniform}[\theta_1^*, r]$ (under which $\theta_2^* = r$ still holds, as noted above).

Under unanimity, W_k must also secure votes from $N-2$ non-proposing weak states. Setting each W_i 's expected payoff from accepting equal to its rejection payoff (the discounted R2 continuation under the prior) and solving gives the binding offer

$$y_{W_i}(\theta_1^*) = \frac{\beta(\theta_1^* + 1 - 2\alpha r)}{2N}. \quad (32)$$

Substituting into W_k 's expected payoff from cutoff $s \equiv \theta_1^*$:

$$\Pi_{W_k}(s) = \frac{1}{r-1} \int_1^s \left[\theta - \frac{\beta(Ns + N - 2 + 2\alpha r)}{2N} \right] d\theta + \frac{r-s}{r-1} \cdot \frac{\beta[s + (1-2\alpha)r]}{2N}, \quad (33)$$

where the first term is the proposer's surplus when H accepts (types $\theta \leq s$) and the second is the discounted R2 payoff when H rejects (types $\theta > s$). Multiplying by $2N(r-1)$ and collecting:

$$2N(r-1)\Pi_{W_k}(s) = \underbrace{[N - (N+1)\beta]}_{s^2 \text{ coeff.}} s^2 + \underbrace{2\beta}_{s \text{ coeff.}} s + (\text{terms in } \alpha \text{ only}).$$

Since α enters only the constant term, the optimal cutoff is independent of α :

$$\theta_1^* = \begin{cases} r & \text{if } \beta \leq \frac{Nr}{(N+1)r-1}, \\ \min\left(r, \frac{\beta}{(N+1)\beta-N}\right) & \text{otherwise.} \end{cases} \quad (34)$$

When $\beta \leq Nr/[(N+1)r-1]$, the unconstrained maximum of the quadratic is at $s^* \geq r$, so $\theta_1^* = r$

(full pooling). When $\beta > Nr/[(N+1)r-1]$, the interior maximum $s^* = \beta/[(N+1)\beta-N] < r$ binds.

H 's payoff from W 's proposal. Given cutoff θ_1^* , types $\theta \leq \theta_1^*$ receive $\beta(\theta_1^* + x)/N$ and types $\theta > \theta_1^*$ receive $\beta(\theta + x)/N$. Integrating over Uniform $[1, r]$:

$$\begin{aligned} E_\theta[V_H^{R1} \mid W\text{-prop}, U] &= \frac{\beta}{N(r-1)} \left[(\theta_1^* - 1)(\theta_1^* + x) + \frac{r^2 - (\theta_1^*)^2}{2} + (r - \theta_1^*)x \right] \\ &= \frac{\beta}{N} \left[\frac{(\theta_1^* - 1)^2}{2(r-1)} + \frac{1+r}{2} + x \right]. \end{aligned} \quad (35)$$

The term $(\theta_1^* - 1)^2/[2(r-1)]$ is the screening rent from R1 pooling. It is zero when $\theta_1^* = 1$ and maximal at $\theta_1^* = r$, giving $(r-1)/2$. This term is absent under majority. Verification: at $\theta_1^* = r$, equation (35) gives $\beta(r+x)/N = \beta V_H^{R2}(r, U)$.

H proposes under majority (probability $1/N$). H buys $q-1$ votes at $\beta E_\theta[V_W^{R2}(\theta, M)] = \beta(1-\alpha)(1+r)/(2N)$:

$$E_\theta[V_H^{R1}(\theta, H\text{-prop}, M)] = \frac{1+r}{2} \left[1 - \frac{(q-1)\beta(1-\alpha)}{N} \right]. \quad (36)$$

W proposes under majority (probability $(N-1)/N$). W excludes H from the winning coalition; H captures $\alpha\theta$. Expected payoff: $\alpha(1+r)/2$.

C.4 Payoff comparison and dominance threshold

Combined R1 payoff under unanimity. Weighting H -proposes by $1/N$ and W -proposes by $(N-1)/N$, and using $\alpha r + x = N\alpha r$:

$$E_\theta[V_H^{R1}(U)] = \frac{1+r}{2N} + \frac{(N-1)\beta}{N^2} \left[\frac{(\theta_1^* - 1)^2}{2(r-1)} + N\alpha r \right]. \quad (37)$$

Combined R1 payoff under majority. With $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ and $C_{\text{out}} \equiv N(N-1)\alpha$:

$$E_\theta[V_H^{R1}(M)] = \frac{(1+r)(N + C_{\text{out}} - C_{\text{buy}})}{2N^2} = \lambda_M \cdot \frac{1+r}{2}, \quad (38)$$

where $\lambda_M = (N + C_{\text{out}} - C_{\text{buy}})/N^2$, as in the binary model (Appendix B.1).

Payoff difference. The difference $D \equiv E_\theta[V_H^{R1}(U)] - E_\theta[V_H^{R1}(M)]$ is

$$D = \frac{1}{2N^2} \left[\frac{(N-1)\beta(\theta_1^* - 1)^2}{r-1} + (1+r)C_{\text{buy}} - C_{\text{out}}(1+r-2\beta r) \right]. \quad (39)$$

D is affine in α . Expanding C_{buy} and C_{out} :

$$2N^2 D = \underbrace{\frac{(N-1)\beta(\theta_1^* - 1)^2}{r-1} + (1+r)\beta(q-1)}_{\equiv A > 0} - \alpha \underbrace{[(1+r)\beta(q-1) + N(N-1)(1+r-2\beta r)]}_{\equiv \Lambda}.$$

The threshold $D = 0$ gives the dominance boundary:

$$\alpha_{\text{cont}}^*(N, \beta, r) = \frac{A}{\Lambda} = \frac{\frac{(N-1)\beta(\theta_1^* - 1)^2}{r-1} + (1+r)\beta(q-1)}{(1+r)\beta(q-1) + N(N-1)(1+r-2\beta r)}. \quad (40)$$

When $\Lambda > 0$, unanimity dominates ($D > 0$) for $\alpha < \alpha_{\text{cont}}^*$. When $\Lambda \leq 0$, unanimity dominates for all $\alpha \in (0, 1/r)$.

Structure of D . The three components have transparent interpretations. The screening-rent term $(N-1)\beta(\theta_1^* - 1)^2/(r-1) \geq 0$ captures the overpayment to low types when W pools in R1 under unanimity; it is absent under majority. The vote-buying term $(1+r)C_{\text{buy}} > 0$ reflects H 's cheaper coalition formation under majority ($q-1$ votes instead of $N-1$). The outside-option term $-C_{\text{out}}(1+r-2\beta r)$ captures the net effect of H 's bilateral alternative: under majority, H captures $\alpha\theta$ when excluded; under unanimity, H captures a pooling offer. The sign depends on β and r .

Comparison with the binary-type threshold. The binary-type threshold (Theorem 1) is

$$\alpha^* = \frac{\beta(q-1)}{N(N-1)(1-\beta) + \beta(q-1)}.$$

Limit $r \rightarrow 1^+$. As $r \rightarrow 1$, the type space degenerates: $\theta_1^* \rightarrow 1$, the screening rent vanishes, and

direct computation gives

$$\alpha_{\text{cont}}^* \rightarrow \frac{\beta(q-1)}{\beta(q-1) + N(N-1)(1-\beta)} = \alpha^*.$$

The two thresholds coincide: the continuous model recovers the binary result when the type space becomes trivial.

For $r > 1$. The numerator A of α_{cont}^* exceeds the (scaled) numerator of α^* by the screening-rent term $(N-1)\beta(\theta_1^* - 1)^2/(r-1) > 0$. A grid search over $N \in \{3, 5, 7, 10, 20, 50, 100, 164\}$, $\beta \in \{0.10, 0.15, \dots, 0.99\}$, $r \in \{1.01, 1.1, 1.5, 2, 3, 5, 10\}$ (1,008 combinations) confirms

$$\alpha_{\text{cont}}^* \geq \alpha^* \quad \text{for all parameter values,}$$

with equality only at $r = 1$. The minimum ratio is $\alpha_{\text{cont}}^*/\alpha^* = 1.010$ (at $r = 1.01$). Unconditional dominance ($\Lambda \leq 0$, implying $D > 0$ for all $\alpha \in (0, 1/r)$) occurs in 32% of the grid. Analytical formulas are verified against Monte Carlo simulation (10^6 draws per parameter combination) to relative error below 5×10^{-4} .

The binary model is therefore the most conservative specification for the mechanism: continuous types amplify the screening rent, enlarging the parameter region where unanimity dominates majority.